EE-4002: Introduction to Digital Signal Processing

HW1

Deadline: March 27, 2025

Question 1

An LTI system has the frequency response

$$H(e^{j\omega}) = \frac{1 - 1.25e^{-j\omega}}{1 - 0.8e^{-j\omega}} = 1 - \frac{0.45e^{-j\omega}}{1 - 0.8e^{-j\omega}} \tag{1}$$

(a) Specify the difference equation that is satisfied by the input $x[n]$ and the output $y[n]$.

(b) Use one of the above forms of the frequency response to determine the impulse response $h[n]$.

(c) Show that $|H(e^{j\omega})|^2 = G^2$, where G is a constant. Determine the constant G. (This is an example of an allpass filter.)

(d) If the input to the above system is $x[n] = cos(0.2\pi n)$, the output should be of the form $y[n] = Acos(0.2\pi n + \theta)$. What are A and θ?

Question 2

The autocorrelation sequence of a signal $x[n]$ is defined as

$$R_x[n] = \sum_{k=-\infty}^{\infty} x^*[k]x[n+k] \tag{2}$$

(a) Show that for an appropriate choice of the signal $g[n]$, $R_x[n] = x[n] * g[n]$, and identify the proper choice for $g[n]$.

(b) Show that the Fourier transform of $R_x[n]$ is equal to $|X(e^{j\omega})|^2$.